

VoLearn: An Operable Motor Learning System with Auditory Feedback

Chengshuo Xia
Keio University
Yokohama, Japan
csxia@keio.jp

Xinrui Fang
Keio University
Yokohama, Japan
xinrui.fang@keio.jp

Yuta Sugiura
Keio University
Yokohama, Japan
sugiura@keio.jp

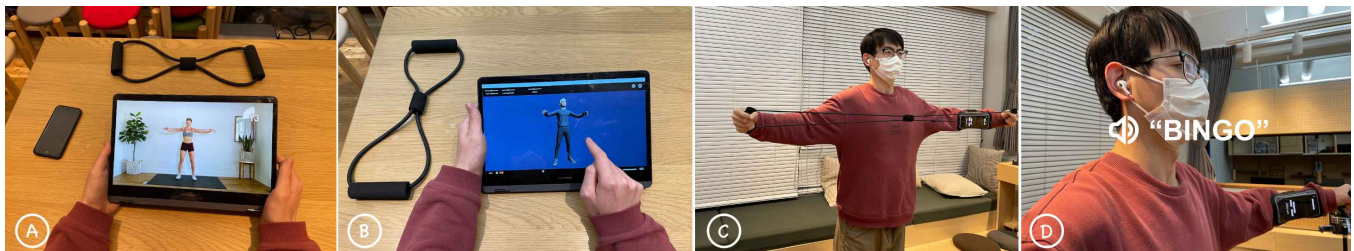


Figure 1: *VoLearn* helps the user in motor learning through operating the motion animation that is converted from video motion source. (a) The desired motion video. (b) The user operates our system with interested function to generate the motion instruction template data. (c) The user wears the smartphone during the motion study and listen to the audio pitch. (d) Once the user reaches the correct position, an auditory cue is generated.

ABSTRACT

Previous motor learning systems rely on a vision-based workflow both from feed-forward and feedback process, which limits the application requirement and scenario. In this demo, we presented a novel cross-modal motor learning system named *VoLearn*. The novice is able to interact with desired motion through a virtual 3D interface and obtain the audio feedback based on a personal smartphone. Both interactivity and user-accessibility of the designed system contribute to a wider range of applications and reduce the limitations in the applied space as well.

CCS CONCEPTS

• **Computer systems organization** → **Embedded systems**; *Redundancy*; Robotics; • **Networks** → Network reliability.

KEYWORDS

Cross-modality, feedback, motor learning, virtual avatar

ACM Reference Format:

Chengshuo Xia, Xinrui Fang, and Yuta Sugiura. 2021. VoLearn: An Operable Motor Learning System with Auditory Feedback. In *The Adjunct Publication of the 34th Annual ACM Symposium on User Interface Software and Technology (UIST '21 Adjunct)*, October 10–14, 2021, Virtual Event, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3474349.3480186>

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UIST '21 Adjunct, October 10–14, 2021, Virtual Event, USA

© 2021 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-8655-5/21/10.

<https://doi.org/10.1145/3474349.3480186>

1 INTRODUCTION

Traditional motor learning systems normally adopt the vision information to a large extent, which maintains the intuitive and informative characteristic [1, 2, 5]. However, the high requirement of vision-tracking devices and limited usage space restrict the motor learning system to be pervasively developed.

In this demo, we presented a novel motor learning system named *VoLearn*, which keeps the feasibility of interacting between the novice and desired motion. The system transferred the conventional studied tutorial video into virtual 3D motion. The user can then operate the motion by controlling the speed, amplitude, and motion combination. Subsequently, with the output template data of desired body segment, we designed an error-based feedback system with an auditory cue to help guide the user conduct the correct motion. The user can employ a consumer device (i.e., smartphone) as a wearable device to train and practice the desired motion according to comprehensive audio feedback.

2 OPERATING THE MOTION

2.1 Generating an operable 3D human motion

In our work, we first leveraged the *Detectron2* model to extract the 2D pose of desired motion in the video frame by frame. And then adding 3D information through the *VideoPose3D* to generate the 3D body skeleton position in a local reference frame [4]. The obtained skeleton is then applied to an avatar through the inverse kinematic tool.

2.2 Motion Manipulation

As the initial motion video is converted into a 3D virtual motion, it is likely to access the key joint movement data and be applied to a humanoid avatar. Therefore, we created an interface to support

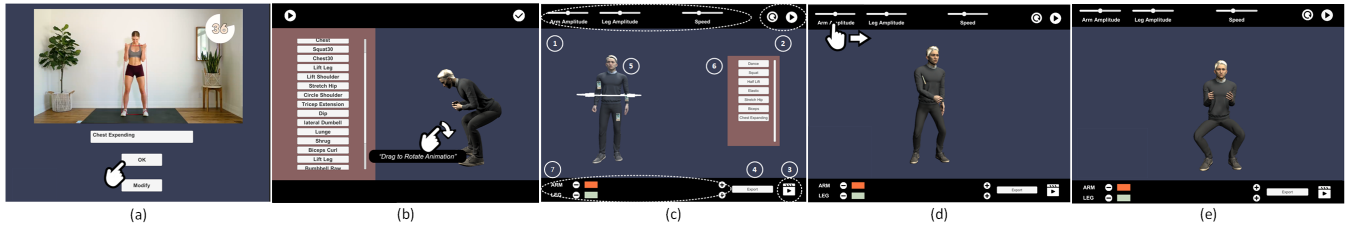


Figure 2: The motion operation interface. (a) The input interface for uploading the desired tutorial video. (b) The management library which stores the before transferred motion. (c) The operable interface with several functions. 1. Is the operation tool including adjustment of motion amplitude and speed. 2. Is the restart and play button. 3. Is the button for viewing the initial RGB video. 4. Is the output button. 5. Is the virtual avatar. 6. Is the motion management area. 7. Is the frequency adjustment tool. (d) An example of adjusting the arm movement amplitude. (e) An example of combining two separate motions of the upper and lower body.

the relevant operation on the transferred motion. Figure 2 presents several interfaces of *VoLearn* system.

We developed the functionality that supports changing the amplitude of motion. It can help to minimize the error of the virtual motion output but also allow the user to adjust the motion according to the consideration of self situation. Additionally, a speed-adjustment tool is also designed to help the novice understand the motion conduction clearly. It can be seen as an alternative of a playback speed functionality during watching a video. Another interface for motion management and combination is designed to combine the upper body motion with lower body motion together to produce a new whole body motion according to the user's idea.

3 FEEDBACK OF AUDITORY PITCH AND CUE

3.1 Virtual template motion data

After the virtual motion is created by the users, the corresponding movement information of the body segment can be derived. A virtual sensor is designed to output the relevant moving data from the 3D virtual motion. Regarding the modality of captured virtual signal, we adopted the rotation data of the joint. Figure 3 shows the virtual smartphone design. The user can freely select the desired body parts to guide and correct the movement of the specific body segment.

3.2 Error based Auditory Feedback

As the motion is expressed in three dimensions, to provide an effective feedback, we converted the 3D motion to an axis-of-effect (AoE) motion [3]. It means that we only compare one axis angle variation and instruct the user to learn the motion based on one axis angle trajectory. According to each three-axis angle variance, the axis of greatest difference is determined as the main compared axis.

The audio feedback is generated from motion direction judgment and the difference between the real angle data and the template data. When the user is conducting the motion, the pitch of the reminder voice (like 'Di Di Di') is going to be increased when the difference is decreasing. Once the extreme value from the template data series has been reached it means that a correct motion is performed. The system is going to produce a sound like 'Bingo' to set an indication.

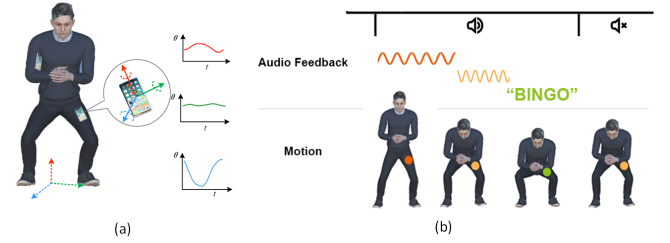


Figure 3: The demonstration of virtual motion data and audio based feedback. (a) The virtual sensor attached on the left upper leg and outputs the rotation data. (b) The illustration of auditory feedback operation with pitch and cue.

After that, the sound disappears and then the user completes one time study.

4 APPLICATION

In this demo, we presented several examples to show the study of motion with the *VoLearn* system. The system supporting process the initial RGB video from the website or device-taken. The *VoLearn* system is developed based on Unity 3D and the real wearable sensor with auditory feedback is designed on an iPhone X.

5 CONCLUSION

In this demo, we presented a novel motor learning system, *VoLearn*. The system is designed from end-to-end. We converted the traditional 2D tutorial video into an operable 3D motion to interact with the novice. To better fit wider application scenarios, an auditory feedback system is implemented based on the personal smartphone. With our developed processing flow and functions, we envision that our system may help more people in motor learning, especially in customized learning.

ACKNOWLEDGMENTS

This work was supported by JST PRESTO Grant Number JPMJPR17J4 and JST AIP-PRISM Grant Number JPMJCR18Y2.

REFERENCES

- [1] Fraser Anderson, Tovi Grossman, Justin Matejka, and George Fitzmaurice. 2013. YouMove: enhancing movement training with an augmented reality mirror. In *Proceedings of the 26th annual ACM symposium on User interface software and technology*. 311–320.
- [2] William Delamare, Thomas Janssoone, Céline Coutrix, and Laurence Nigay. 2016. Designing 3D gesture guidance: visual feedback and feedforward design options. In *Proceedings of the International Working Conference on Advanced Visual Interfaces*. 152–159.
- [3] Yousef Kowsar, Masud Moshtaghi, Eduardo Velloso, Lars Kulik, and Christopher Leckie. 2016. Detecting unseen anomalies in weight training exercises. In *Proceedings of the 28th Australian Conference on Computer-Human Interaction*. 517–526.
- [4] Dario Pavlo, Christoph Feichtenhofer, David Grangier, and Michael Auli. 2019. 3d human pose estimation in video with temporal convolutions and semi-supervised training. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 7753–7762.
- [5] Richard Tang, Xing-Dong Yang, Scott Bateman, Joaquim Jorge, and Anthony Tang. 2015. Physio@ Home: Exploring visual guidance and feedback techniques for physiotherapy exercises. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*. 4123–4132.